Galois module structure of units of rings of integers of (C_4) number fields

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AGQ-CDT

Integral representations

- Let G be a finite group.
- Usually, G acts on a vector space V over some field K, giving a map
$$G \to \mathrm{GL}(V).$$
- G can also act on a module over some integral domain R, e.g. $R = \mathbb{Z}$.
- However, some properties no longer hold.
- For example, Maschke's theorem no longer holds (failure of decomposition into irreducibles).
- Therefore, we consider indecomposable representations instead.

Indecomposable representations for $G = C_4$

There are 9 indecomposable integral representations of C_4 :

- $[1],\ [-1]$,
- $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}$,
- $\begin{bmatrix} 0 & -1 & 1 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}, \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$,
- $\begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}, \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 0 & -1 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}, \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$.

Note: the situation gets simpler when considering representations coming from number theory.

Setup: Unit groups

- Let K be a number field.
- Assume K is Galois and $\mathrm{Gal}(K|\mathbb{Q}) \simeq C_4$.
- Consider its ring of integers $\mathcal{O}_K$, which is the integral closure of $\mathbb{Z}$ in K.
- Let $\mathcal{O}_K^\times$ be the unit group. Let r_1 be the number of real embeddings $K \hookrightarrow \mathbb{R}$ and r_2 the number of pairs of complex embeddings $K \hookrightarrow \mathbb{C}$ that are not real. Then $[K : \mathbb{Q}] = r_1 + 2r_2$.
- Dirichlet Unit Theorem:

$$\mathcal{O}_K^\times \simeq \mu(K) \times \mathbb{Z}^{r_1+r_2-1}.$$

- Example: for $K = \mathbb{Q}(\zeta_5)$, $\mathcal{O}_K^\times \simeq \mu_{10} \times \mathbb{Z}$.

Algorithm

- The input of the algorithm is a number field K with $\mathrm{Gal}(K|\mathbb{Q}) = C_4$, and the output is a decomposition of the integral representation $\mathcal{O}_K^\times$ into indecomposable representations.
- Since $r_1 + 2r_2 = 4$, we have two options: either $r_2 = 2$ and $r_1 = 0$ (totally imaginary, so rank 1 and only one option) or $r_1 = 4$ and $r_2 = 0$ (totally real and rank 3, so more options).
- Step 1: take the number field K and return a matrix (representing the action of the Galois group on $\mathcal{O}_K^\times$).
- Step 2: compare that matrix to the ones associated to indecomposable representations, and check if they are similar (do it after base-changing to $\mathbb{Q}$ and $\mathbb{F}_2$ for simplicity).